COHOMOLOGY OF A SHEAF SUPPORTED ON A FINITE-PRESENTATION CLOSED SUBSPACE IS POWER-TORSION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1956-1967 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 1969-1991, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_spaces-pushouts.json (pb=1217, stmt 1956-1967, proof 1969-1991).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05E6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1991–1991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Z0: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1991–1991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 071Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1991–1991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $i : Z \to X$ be a closed immersion of finite presentation. Let $Q \in D_{QCoh}(\mathcal{O}_X)$ be supported on $|Z|$. Let $\overline{x}$ be a geometric point of X and let $I_{\overline{x}} \subset \mathcal{O}_{X,\overline{x}}$ be the stalk of the ideal sheaf of Z. Then the cohomology modules $H^n(Q_{\overline{x}})$ are $I_{\overline{x}}$-power torsion (see More on Algebra, Definition 05E6).*

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose an affine scheme U and an étale morphism $U \to X$ such that $\overline{x}$ lifts to a geometric point $\overline{u}$ of U. Then we can replace X by U, Z by $U \times_X Z$, Q by the restriction $Q|_U$, and $\overline{x}$ by $\overline{u}$. Thus we may assume that $X = \mathrm{Spec}(A)$ is affine. Let $I \subset A$ be the ideal defining Z. Since $i : Z \to X$ is of finite presentation, the ideal $I = (f_1, \ldots, f_r)$ is finitely generated. The object Q comes from a complex of A-modules $M^\bullet$, see Derived Categories of Spaces, Lemma 071Q and Derived Categories of Schemes, Lemma 06Z0. Since the cohomology sheaves of Q are supported on Z we see that the localization $M^\bullet_f$ is acyclic for each $f \in I$. Take $x \in H^p(M^\bullet)$. By the above we can find n_i such that $f_i^{n_i} x = 0$ in $H^p(M^\bullet)$ for each i. Then with $n = \sum n_i$ we see that I^n annihilates x. Thus $H^p(M^\bullet)$ is I-power torsion. Since the ring map $A \to \mathcal{O}_{X,\overline{x}}$ is flat and since $I_{\overline{x}} = I\mathcal{O}_{X,\overline{x}}$ we conclude. $\square$